Deakin University

Engineering AI Solutions

OnTrack Submission

Task 5.1P

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2026/04/16 16:39

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April 16, 2026

SIT788 – Engineering AI Solutions

Task 5.1P: Object Detection

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1. Scenario Definition

The selected real-life scenario to do this task is Autonomous Driving. Autopiloted vehicles have to detect and categorize an extensive variety of things in real-time, such as pedestrians, traffic, other vehicles, traffic lights, road signs and bicycles, under extremely changing environmental factors like lighting, weather, and traffic congestion.
A failure to recognize a pedestrian on the road or to recognize a stop sign is life-threatening in autonomous driving systems. AI-driven systems are also completely dependent on the accuracy and reliability of their perception models, unlike human drivers who rely on intuition and assessing the situation. This not only makes object detection a technical necessity, but a basic safety requirement.
Real-world deployment requires the capability to adjust detection thresholds, to deal with cluttered urban scenes, and to be robust under degraded visual conditions, like fog, rain, or driving at night. This assignment will assess the performance of the state-of-the-art pre-trained models in these diverse conditions.

2. Threshold Tuning Analysis

In Image 1 there is a clear daytime city street view where there are pedestrians crossing a zebra crossing on the street, vehicles in the background and traffic lights on either side of the road. YOLO26 model (YOLO26n) was tested with three levels of confidence threshold on Hugging Face Spaces.

Threshold	Total Detected	False Positives	False Negatives	Key Observations
0.25 (Low)	10	~1	~2	Traffic light detected at 0.43. Persons, cars, handbag all detected including background objects. One likely false positive from shadow.
0.50 (Medium)	6	~0	~4	Traffic light dropped out (0.43 < threshold). Only high-confidence persons and cars remain. Cleaner but less complete.
0.85 (High)	0	0	~10	Zero objects detected. All objects fell below the 0.85 threshold. Model is effectively blind at this setting.

Table 1: Threshold Tuning Results – Image 1 (Ideal Conditions)

Screenshots – Threshold Testing

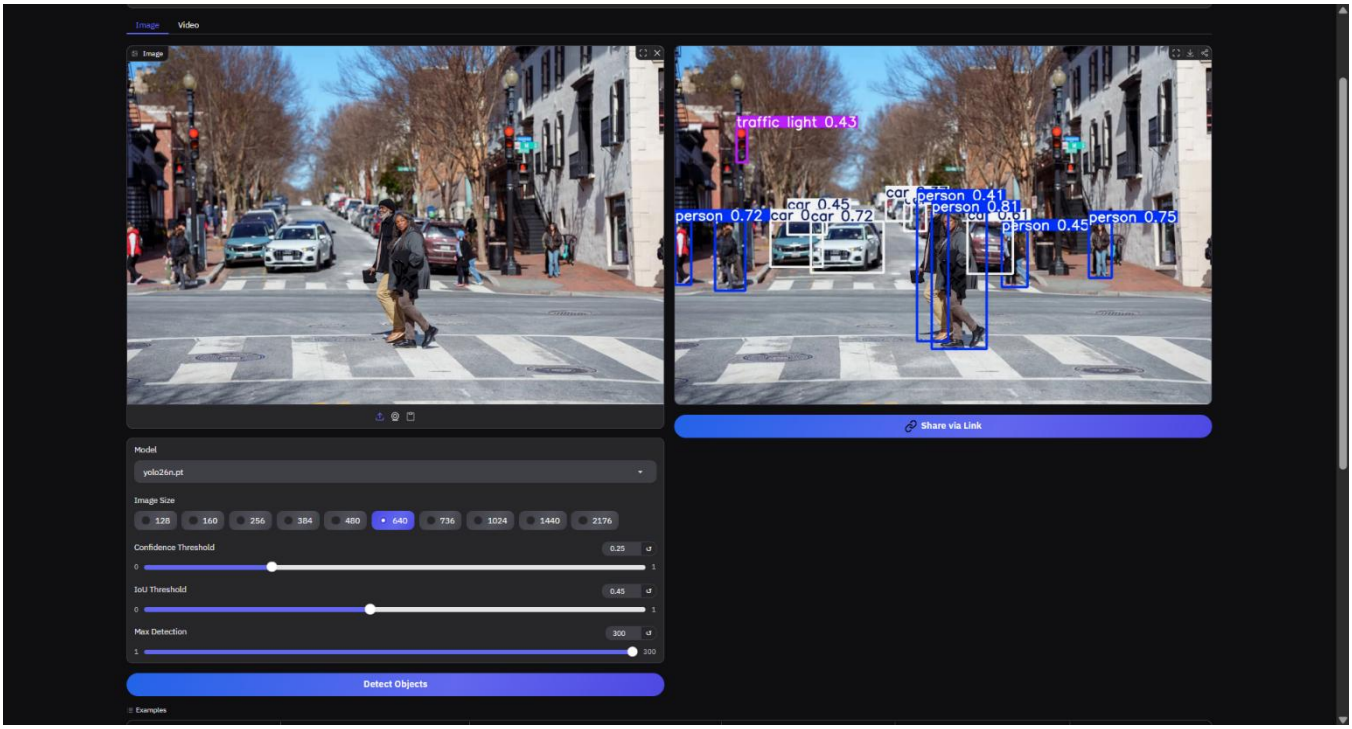

Figure 1: Image 1 at Confidence Threshold 0.25 – 10 objects detected (persons, cars, traffic light)

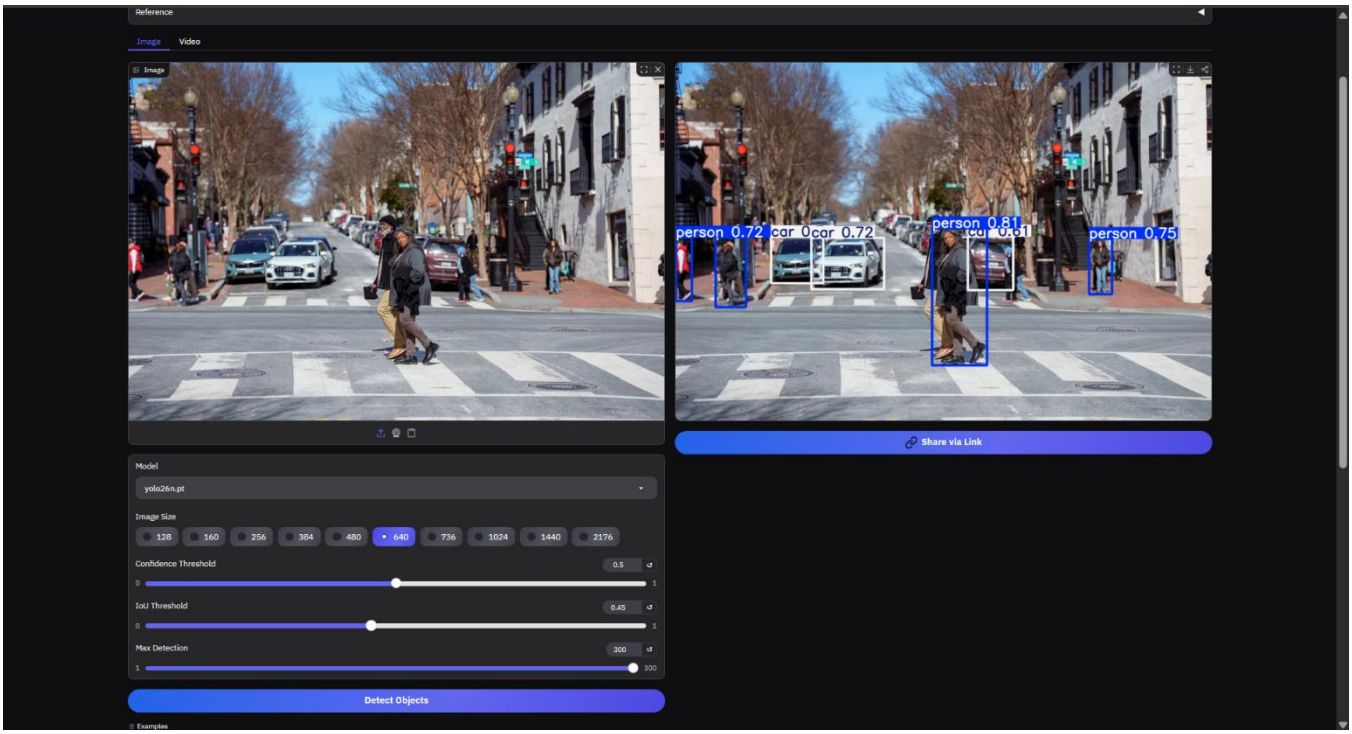

Figure 2: Image 1 at Confidence Threshold 0.50 – 6 objects detected (traffic light dropped out)

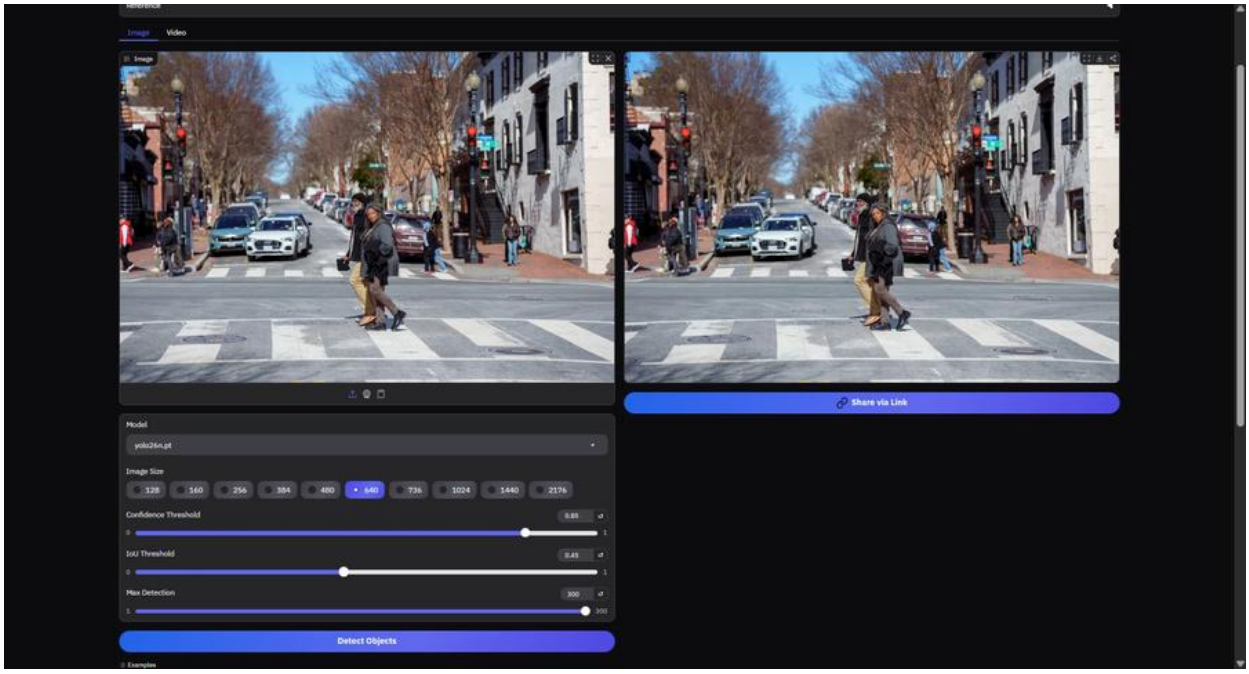

Figure 3: Image 1 at Confidence Threshold 0.85 – 0 objects detected

Reflection – Threshold Tuning

The findings illustrate the underlying trade-offs between false positive and false negative as the confidence threshold is changed. The model at 0.25 identified 10 objects, which capture the greatest content of the scene, yet brings an estimated false positive due to a shadow area, and a traffic light with a marginal confidence value of 0.43. Although this maximizes the recall, the chance of acting on false detections is high in autonomous driving, where false detections may result in unnecessary braking or evasive manoeuvres.

At 0.50, there were no longer 6 detections, and the results were cleaner. Nevertheless, about four real objects were missed which is an increment in false negatives. In the case of an autonomous vehicle, failure to detect a pedestrian or traffic signal is a major safety issue.
The model was unable to identify anything at 0.85, which shows that a very strict limit makes the system blind. The most practical starting point is a balanced threshold of about 0.50 and additional tuning is suggested depending on environmental conditions.

3. Edge Case Hypotheses

Two hard edge-case images were tested with the YOLO26 model at a constant confidence threshold of 0.50. In Image 2, there is a busy city traffic shot, taken at an aerial view with a lot of traffic congestion, and Image 3 shows a vehicle on a foggy, wet road at night.

3.1 Image 2 – Cluttered Scene (YOLO26)

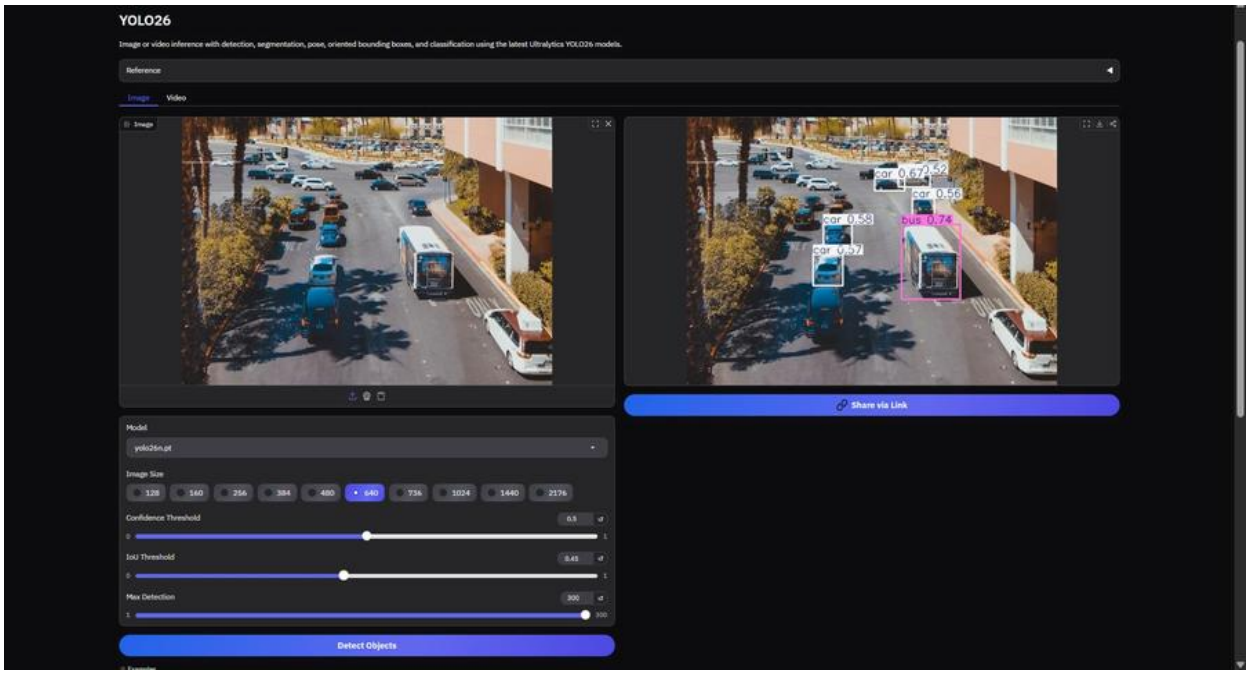

Figure 4: Image 2 (Cluttered Intersection) – YOLO26 at threshold 0.50. Only 6 objects detected from a scene with 15+ vehicles.

Detections: 5 cars (0.52–0.67), 1 bus (0.74). Total: 6 objects.

Failures: Traffic light missed. Unnoticed trucks. ~10 in the background and low-left cluster are completely missed.

The aerial overhead view is the major cause of poor performance of the model on Image 2. Like most of the YOLO-family models, YOLO26 is trained on the COCO dataset, which is mostly ground-level photography. Cars seen in overhead shots create a radically different visual signature - a top-view of the roof of a car has little similarity to the frontal or lateral view which constitutes most training images. Consequently, most vehicles are not recognised in this point of view by the feature extraction mechanisms of the model.

Another reason is inter-object occlusion. The clustering of multiple cars with overlapping bounding regions is observed in the lower-left quadrant. The model has trouble with determining individual object boundaries when objects overlap or are obscured by other moving objects. The dark shadows on the scene additionally decrease contrast on the edges of the objects and worsen the quality of feature maps utilized in the classification. Traffic light failure to be identified can be explained by its small apparent size and location at the edge of the image.

3.2 Image 3 – Degraded Conditions (YOLO26)

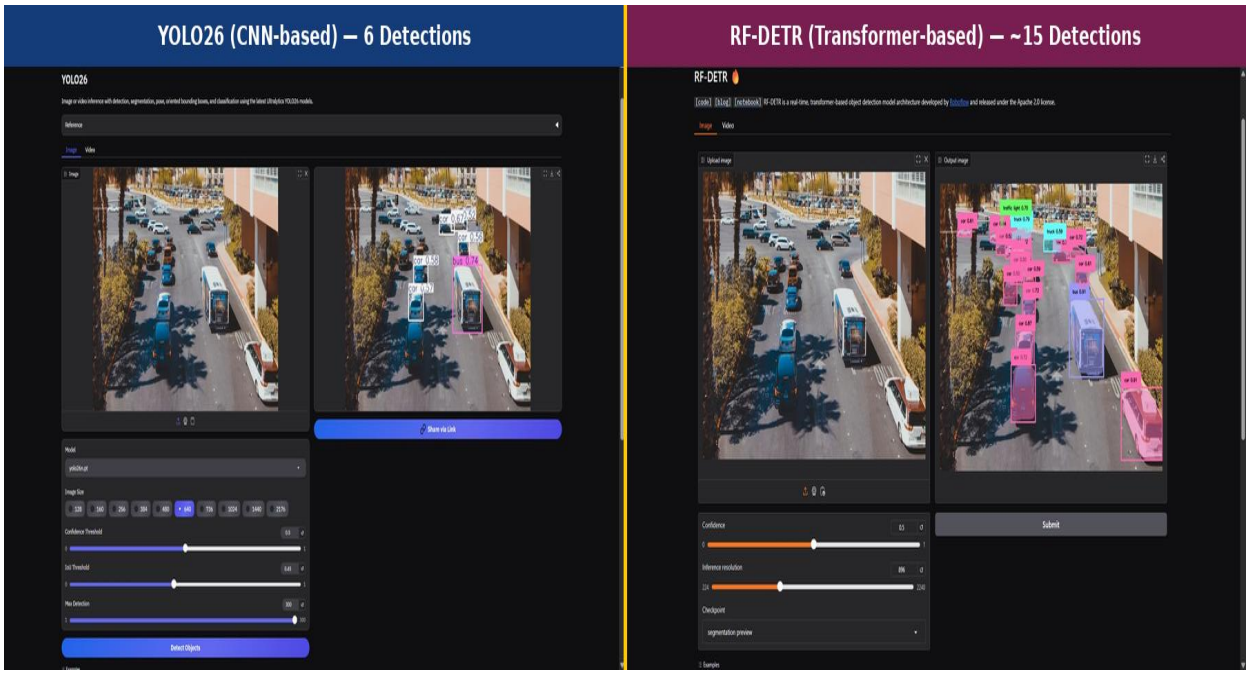

Figure 5: Image 3 (Foggy Night Road) – YOLO26 at threshold 0.50. Only 1 object detected: car (0.83).

Detections: 1 car (0.83). Total: 1 object.

Failures: Not detected road markings. Unclassified headlight glare. Immediate surrounding (trees, barriers) totally overlooked. Bounding box only includes the front part of the vehicle.

The ability of Image 3 to almost lose all the scene information can be explained by the fact that the pixels have been degraded because of heavy fog and low-light conditions. The fog diffuses light evenly over the picture, decreasing the contrast and colour difference between objects and backgrounds. This dwarfs high-frequency edge details on which convolutional neural networks depend to detect object boundaries. The vehicle was only visible by the model due to the headlights that provided a high local contrast on the dark road surface.

That the incomplete bounding box only covers the front of the vehicle is compatible with this description - the rear of the SUV was obscured by fog, so the model did not have enough pixel data to extrapolate its capture to the entire size of the vehicle. The lane markings were not identified because the lane lines have not been specified as the object classes in the COCO dataset. This is where the major weakness of typical pre-trained models lies in autonomous driving in unfavourable weather conditions.

4. Benchmarking Comparison

Image 2 (the disorganized overhead intersection) was chosen as the most complex image to be benchmarked because YOLO26 only detected 6 out of 15+ objects. The other model (RF-DETR (Real-time Detection Transformer by Roboflow) was run on the same image with the same confidence threshold of 0.50 using Hugging Face Spaces.

4.1 Model Architecture Comparison

Attribute	YOLO26	RF-DETR
Architecture	CNN-based (convolutional)	Transformer-based (attention)
Detection Method	Anchor-free, NMS-free single-pass	End-to-end set prediction with DINOv2
Training Data	COCO (ground-level)	COCO + diverse augmentations
Hugging Face Space	nicehuster/yolo26	Roboflow/RF-DETR

Table 2: Model Architecture Overview

4.2 Side-by-Side Visual Comparison

The following screenshots show both models applied to Image 2 (Cluttered Intersection) at confidence threshold 0.50.

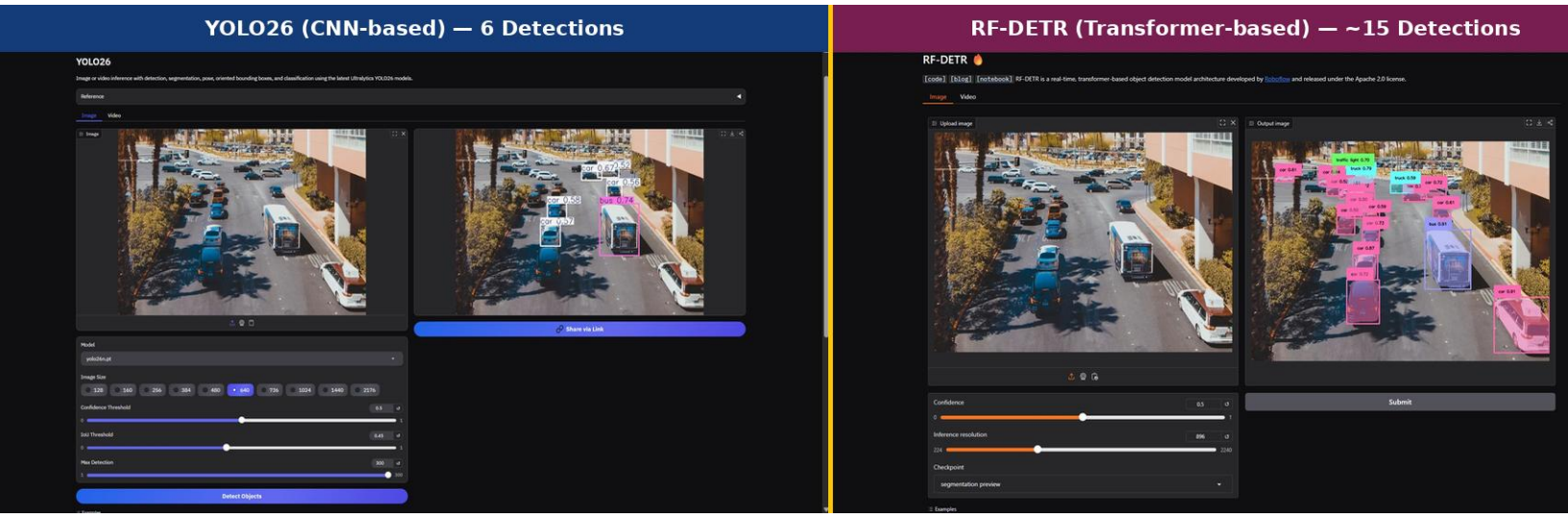

Figure 6: Side-by-Side Comparison – YOLO26 (left, CNN-based, 6 detections) vs RF-DETR (right, Transformer-based, ~15 detections) on Image 2 (Cluttered Aerial Intersection) at confidence threshold 0.50.

4.3 Results Comparison Table

Metric	YOLO26	RF-DETR	Observation
Total objects detected	6	~15	RF-DETR detected ~2.5x more objects
Traffic light detected	No	Yes (0.70)	RF-DETR identified the traffic signal
Trucks detected	No	Yes (0.79, 0.59)	YOLO26 missed trucks entirely
Bus confidence	0.74	0.91	RF-DETR more confident on same object
Aerial view handling	Poor	Good	Transformer attention handles non-standard angles better

Table 3: Benchmarking Results – YOLO26 vs RF-DETR on Image 2

4.4 Critical Reflection & Recommendation

The benchmarking data indicates that RF-DETR has a clear and significant performance gain over YOLO26 on the most difficult edge-case image. RF-DETR was able to find around 15 objects with YOLO26 finding only 6, it correctly recognized trucks and a traffic light which YOLO26 failed to find at all and scored higher on shared detections like the bus (0.91 vs 0.74).

This performance difference can be explained by the inherent difference between the architecture of the two models. YOLO26 is based on a CNN backbone, which is effective in regular ground-level images and fails in non-regular ones like the overhead view in Image 2. It depends on learned spatial filters, which are sensitive in the orientation of objects as observed in training, in order to extract features.

In contrast, RF-DETR uses a transformer-based architecture, which has self-attention mechanisms that learn global spatial associations between the entire image. This enables the model to take into account the context of objects in relation to each other, instead of basing it on local patterns of pixels, and is more resistant to changes in perspective and unusual viewpoints.

According to this analysis, RF-DETR is the model architecture that should be used in autonomous driving applications. The fact that it can more easily find more objects with more confidence in more complex, cluttered, non-standard visual scenes is much more robust than YOLO26 to deploy in the real world.

References

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